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Subject - NSPA

Assignment - 1

Q-1)

A-1) Firewall Applications

A firewall is a security system that monitors and controls incoming and outgoing network traffic

Uses :-

- Network Security - Blocks unauthorized access.
- Traffic Filtering - Allows only trusted data packets.
- Prevent Cyber attacks - Stops malware, hackers.

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Real - World Examples —

- Banks use firewalls to protect customer financial data.
- Schools block social media using firewalls.

NGFW (Next - Generation Firewall).

- NGFW is an advanced firewall with extra features like deep inspection and application awareness.

Features —

- Deep packet Inspection (DPI).
- Intrusion Prevention System (IPS).

Real world Examples —

- Corporate networks use NGFW to monitor employee activity.
- Cloud systems use NGFW to protect applications.

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Q.2)

A.2) HTTP vs HTTPS

- HTTP → Not secure (data is plain text).
- HTTPS → Secure (uses encryption via SSL / TLS).

Security Issues –

1) Header Manipulation

Attackers modify HTTP headers to :

- Bypass authentication
- Trick servers

2) Lack of HSTS (HTTP Strict Transport Security).

If HSTS is not enabled :

- Users can be forced to use HTTP instead of HTTPS
- Vulnerable to Man-in-the-Middle (MITM) attacks.

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Q-3) SSL/TLS Handshake Process (with diagram).

Steps :

1) Client Hello

- Client sends request + supported encryption methods

2) Server Hello

- Server selects encryption method and sends certificate.

3) Certificate Verification

- Client verifies server identity.

4) Key Exchange

- Secure key is generated.

5) Secure Connection Established

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Client → Server : Client Hello
Server → Client : Server Hello + Certificate
Client → Server : Key Exchange
Server → Client : Confirmation
→ Secure communication starts

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Project

Section -1:

A-1) Problem

A company had a data breach despite using a firewall.

Why Firewall Failed:

1) Packet Filtering Limitation

- Only checks IP and port, not data content.

2) Stateful Firewall Limitation.

- Tracks sessions but cannot detect advanced attacks.

3) No Application Awareness

- Malicious traffic hidden in allowed application (eg. HTTP).

4) Encrypted Traffic Issue.

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- Cannot inspect HTTPS properly

Why ~~Fira~~ Firewall Failed—

1) Packet Filtering Limitation.

- Only checks IP and port, not data content.

2) Statiful Firewall Limitation.

- Tracks session but cannot detect advanced attacks

3) No Application Awareness

- Malicious Traffic hidden in allowed applications.

4) Encrypted Traffic Issue

- Cannot inspect HTTPs properly.

Improvements using NGFW & Application Firewalls.

1) NGFW solutions.

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- Deep packet inspection
- Detect malware inside traffic

2) Application Firewall (WAF).

Protects web applications specifically.

Functions:

- Blocks SQL injection.
- Stops Cross-site scripting (XSS).

Section -2

Attack Scenario -

- HTTP Header manipulation used
- No HSTS enabled

Attack Explanations:

1) Attacker intercepts HTTP request.
2) Steals sensitive data.

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Problems Identified—

- Use of HTTP instead of HTTPS.
- No HSTS protection.

Preventive Measures:

1) Enable HTTPS Everywhere

- Use SSL/TLS encryption

2) Enable HSTS

- Forces browser to use HTTPS only.

3) Secure Headers

- Use headers like:
- Content-Security-Policy (CSP).
- Secure Cookies.

4) Input Validation.

- Validate all user inputs.

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5) Use Web Application Firewall (WAF).

- Detect and block malicious requests